

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	BS-EMGT
Program Name	Bachelor of Science in Engineering Management
Assessment Year	2024-2025
Program Assessment Coordinator	Stephen Kleps, Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	4
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

This annual report covers activities from August 2024 through July 2025. The BS in Engineering Management (BS-EMGT) operates within the Armour College of Engineering and is administratively supported by the Department of Civil, Architectural, and Environmental Engineering (CAEE). The Program Assessment Coordinator for CAEE is Dr. Stephen M. Kleps, Ph.D., P.E.

Methodology

The Bachelor of Science in Engineering Management program offers an opportunity for students to obtain skills and competencies that are highly relevant and driven by the accelerating development of new technologies in the emerging global economy at the intersection of engineering invention and business administration.

The program's objective is to prepare students to become leaders in the corporate world shaped by innovations in engineering. Students learn fundamentals of science, engineering management, and business administration by concentrating on the development of critical thinking skills directed toward practical problem solving and informed decision-making.

Students completing this program are uniquely positioned to make decisions concerning product process development in ways that combine technical, financial, marketing, human resources, and strategic considerations. Students are prepared to perform economic analyses for new products, evaluate technologies, and assess business processes. Students completing this program will be able to prepare business plans that include financial details, marketing strategies, and design decisions based on target costs and forecasted rate of return on investment capital. The program also includes a business curriculum that focuses on developing organization and management, critical thinking, and entrepreneurship skills.

Each student must select a core engineering specialization from one of the following:

- Aerospace Engineering
- Architectural Engineering
- Biomedical Engineering
- Chemical Engineering
- Civil Engineering
- Computer Engineering
- Computer Science
- Electrical Engineering
- Materials Science and Engineering
- Mechanical Engineering

The program serves an important function within multiple engineering departments within the Armour College of Engineering. It historically serves as a safety net to students that may not have found academic success in their original engineering majors.

Learning objectives for the program include the following:

1. Make decisions concerning product or process development.
2. Perform economic analyses for new products, to evaluate technologies, and to assess business processes.
3. Prepare business plans that include financial details, marketing and design decisions based on project requirements, project goals, quality assurance, scheduling, work breakdown structure, targeted profit and costs, and forecasted rate of return on investment capital.
4. Combine technical, financial, strategic considerations, and human resources such as project team formation, communication skills and personality dynamics, in decision making on engineering projects.

On an annual basis, the CAEE Program Assessment Coordinator reviews student performance in select courses that cover content that matched the learning outcomes. Table 1 provides a list of which of the 1-4 LOs are covered – and importantly, assessed – in required courses in the EMGT program. Highlighted cells show which LOs will be assessed in which courses, with emphasis on (i) primarily summative assessments of each LO, (ii) formative assessment of some LOs, and (iii) breadth of coverage in the basic areas of the program.

Table 1. Mapping of learning objectives (LOs) coverage/assessment in the EGMT program

Course ↓	Learning goals →	1) Decision Making on Products and Processes	2) Economic Analysis of Processes	3) Business Planning to meet Capital Goals	4) Decision Making on Engineering Projects
BUS 211 Financial Accounting		A	A	I	I
BUS 212 Managerial Accounting		A	A	I	D
BUS 301 Organizational Behavior		D	D	D	A
BUS 305 Operation and Supply Chain Analytics		A	A	I	I
BUS 361 Topics in Entrepreneurship		I	I	I	I
BUS 371 Marketing Fundamentals		I	I	A	I
CAE 312 Engineering Systems Analysis		I	I	I	A
COM 421 Technical Communication		A		I	
ECON 211 Introduction to Economics		I	I	I	A
ECON 312 Intermediate Macroeconomics		I	I	I	A
ECON 382 Business Economics		I	A	I	I
ECON 423 Economics of Capital Investments		I	I	A	A
EMGT 406 Entrepreneurship & Intellectual Property Management		I	I	A	I
EMGT 470 Project Management		D	D	A	A

A = Assessed (i.e. strongly supported)

D = Developed (i.e. supported)

I = Introduced (i.e. minimally supported)

Yellow = Course to be used for assessing attainment of individual LOs

Results

Data

Of the 25 students that were identified as being enrolled in the BS-EMGT program in Fall 2024, three students had either just started in the program or otherwise had limited coursework completed, or both. Thus, 22 students within the program were evaluated for program assessment in this cycle. An additional three students transitioned out of the program (either back to their original engineering degree or chose a new degree), but since they took the applicable EMGT coursework during this assessment period, they were included in the 22-student data set. Their concentration breakdown is as follows:

- Aerospace Engineering (1)
- Architectural Engineering (1)
- Biomedical Engineering (2)
- Chemical Engineering (0)
- Civil Engineering (4)
- Computer Engineering (0)
- Computer Science (1)
- Electrical Engineering (1)
- Materials Science and Engineering (0)
- Mechanical Engineering (9)
- Transitioned to Non-EMGT Degree (3)

Total: (22)

As a primarily safety net program, 41% of the students in the program in Fall 2024 had come from the B.S. in Mechanical Engineering major, while the remainder came from aerospace, architectural, biomedical, computer science, electrical engineering or civil engineering.

Table 2 shows the summary results of the program assessor evaluating the student performance in the courses that match up to the LOs for Fall 2024 and Spring 2025. The rating was based on three assessment categories: does not meet expectations, meets expectations, or proficient (i.e., exceeds expectations). Only the percentages of students meeting or exceeding expectations are shown for clarity. Artifacts in each course are stored by instructors and are available upon request.

Table 2. Percentages of students meeting (M) or exceeding (E) expectations (Fall 2024 & Spring 2025)

Course / LG	1	2	3	4
BUS 211	100	100		
BUS 212	100	100		
BUS 301				100
BUS 305	100	100		
BUS 361				
BUS 371			100	
CAE 312				50
COM 421	100			
ECON 211				100
ECON 312				100
ECON 382		75		
ECON 423			75	75
EMGT 406			100	
EMGT 470			100	100
Average for Program	100	94	94	88

The average percentage of students in the program meeting or exceeding expectations for each learning objective (LO) from all instructor assessments across all courses was at or above 70% for all LOs for Fall 2024 and Spring 2025. Additionally, Figure 1 shows distributions of proficiency levels for LOs #1-4 aggregated across all courses for this academic year.

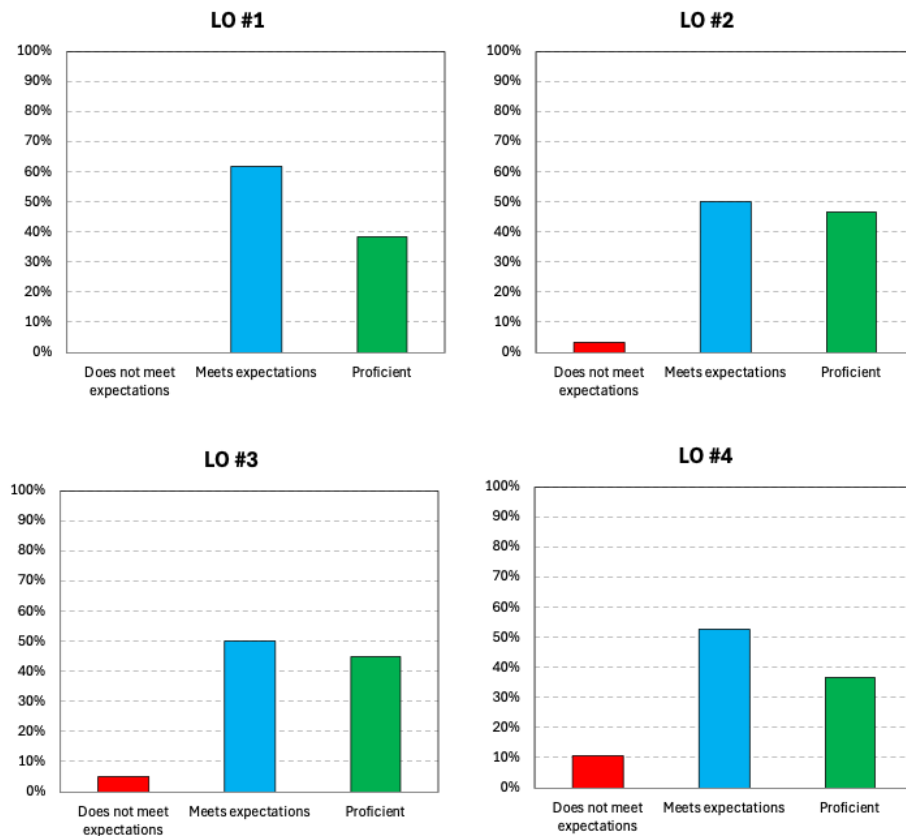


Figure 1. Distributions of proficiency levels for LOs #1-4 aggregated across all courses in this cycle

Description of findings

Discussion of Results by Learning Objective

1) Make decisions concerning product or process development.

The average percentage of students meeting or exceeding expectations across all courses for LO 1 was above 70% for all courses in this academic year. There were no concerns on this LO in this academic year.

2) Perform economic analyses for new products, to evaluate technologies, and to assess business processes.

The average percentage of students meeting or exceeding expectations across all courses for LO 2 was above 70% for all courses in this academic year. There were no concerns on this LO in this academic year.

3) Prepare business plans that include financial details, marketing strategies, and design decisions based on target costs and forecasted rate of return on investment capital.

The average percentage of students meeting or exceeding expectations across all courses for LO 3 was above 70% for all courses in this academic year. There were no concerns on this LO in this academic year.

4) Combine technical, financial, marketing, human resources, and strategic considerations in decision making on engineering projects.

The average percentage of students meeting or exceeding expectations across all courses for LO 4 was above 70% for all courses except for CAE 312 in this academic year. Given that there was a limited data set of only two students in CAE 312, there were no concerns on this LO in this academic year. One student met the expectations, and one did not. In addition, the overall average for the LO was well above the goal of 70%.

Prior CI plan impact

There was one action item from the AY 2023-2024 that was discussed and resolved this year. That action item was that by the end of this year, the CAEE Department Chairman as well as both the Associate Chair of Undergraduate Affairs for CAEE and the Associate Chair of Graduate Affairs will meet to review and confirm the reaffirmation (or not) the PEO and four LOs for the program. As previously discussed in this report, starting in April 2025, the BS-EMGT program assessment coordinator held meetings within the department to discuss updating the Learning Goals for the future academic years. The meetings substantively reaffirmed the LOs except for minor modifications to LOs 3 and 4. We also moved to the university's standard template for assessment reporting in this cycle.

This cycle's CI plan

Given that no averages for any of the four LOs was below our department's 70%, no action items have been introduced in this academic year in response to assessment of outcomes and engagement with program constituents.

These will then be shared with the faculty of the CAEE department and relevant faculty and staff in the Armour College of Engineering, including professional advisors in the Armour Academy and the associate dean for undergraduate affairs.

This engineering management program continues to serve an important function within multiple engineering departments within the Armour College of Engineering. It continues to serve as a safety net to students that may not have found academic success in their original engineering majors, in that the engineering management program still offers these diverse learners a pathway to a useful degree.

APPENDIX

HLC Meeting Minutes for CAEE Department

Topic: HLC Meeting Minutes for Discussion on the Learning Goals (LG's) used in the BS-EMGT program

Date: April 22, 2025, 4:00 pm

Attending:

1. Dr. Brent Stephens, Professor and Chair for CAEE
2. Dr. Steve Kleps, Associate Chair UGSC for CAEE and Program Assessment Coordinator for CAEE

Discussion:

This meeting was held as a self-imposed action item from our CAEE Department's annual HLC report for academic year 2023-2024. In that report, Dr. Kleps, the report author and Program Assessment Coordinator for CAEE, requested a meeting with the department chairman, Dr. Stephens and the Associate Chair for the Graduate Studies Committee for CAEE, Dr. Mehdi Modares. The meeting had two purposes. The first purpose of the meeting was to review the four LG's for the EMGT program and either reaffirm them or update them. The second purpose was to determine if EMGT 470, Project Management, should become a required course in the BS-EMGT program as opposed to an elective within the program.

Drs. Stephens and Kleps agreed that EMGT 470 should become a required course the program. The reasons were that it is an important course within the program, it is taught within our CAEE department so quality control can be more easily maintained, at its core EMGT 470 directly addresses LG's 3 and 4 of our program and most students take the course anyway. In addition, it includes a final project with student deliverables that would allow for a more efficient collection of student artifacts for the program. In a later discussion with Dr. Modares, who could not attend the meeting, he concurred with requiring EMGT 470.

With this decided, Dr. Kleps will propose this update to our program in the Illinois Tech Undergraduate Studies Committee in Spring 2026, hopefully receiving full approval from the University in Spring 2026. This should allow this update to be reflected in Fall 2026.

With regards to the second purpose of the meeting, LG's 1 and 2 were reaffirmed. LG's 3 and 4 were substantially reaffirmed but with minor alterations. These alterations were meant to better reflect the important role that our course EMGT 470 held in the program. In a later discussion with Dr. Modares, he concurred with the updates to the LG's. With three CAEE administrators approving the updates to the LG's, it was decided the last step would be to have the professor of EMGT 470, Ms. Deborah Steimel-Clair, review the LG's to make sure she concurred too. That meeting will occur later this 2025 Spring semester.

Meeting concluded at 4:30 pm.

HLC Meeting Minutes for CAEE Department

Topic: HLC Meeting Minutes for Additional Discussion on the Learning Goals (LG's) used in the BS-EMGT program and EMGT 470 requirement

Date: May 6, 2025, 12:45 pm

Attending:

3. Dr. Steve Kleps, Associate Chair UGSC for CAEE and Associate Teaching Professor
4. Ms. Deborah Steimel-Clair, Associate Teaching Professor, CAEE and Instructor for EMGT 470

Discussion:

This meeting was held as a continuation of a self-imposed action item from our CAEE Department's annual HLC report for academic year 2023-2024. In that report, Dr. Kleps, the report author and Program Assessment Coordinator for CAEE, requested a meeting with the department chairman, Dr. Stephens and the Associate Chair for the Graduate Studies Committee for CAEE, Dr. Mehdi Modares. The meeting had two purposes. The first purpose of the meeting was to review the four LG's for the EMGT program and either reaffirm them or update them. The second purpose was to determine if EMGT 470, Project Management, should become a required course in the BS-EMGT program as opposed to an elective within the program.

That meeting determined that EMGT 470 would be a required course within the EMGT program. It also substantially reaffirmed the LG's except for minor modifications, that Dr.'s Stephens, Kleps and Modares determined that professor of EMGT 470, Ms. Deborah Steimel-Clair, review the modified LG's to make sure she concurred too.

This meeting also had the purpose of confirming that the final project for EMGT 470 satisfied the LG's 3 and 4 for the program. Ms. Steimel-Clair liked the modifications to LG's 3 and 4 and indicated that the modifications fit well into her LG's for the course and more specifically for the final project of the course. She also indicated that her currently grading rubric for the final projects would fit well into the use of the final projects as student artifacts for the course. For thoroughness the original LG's, followed up the modified LG's as approved in the April 22, 2025 meeting and again in this meeting, are included below.

Learning Goals (LGs) Adopted in October 2020

1. Make decisions concerning product or process development.
2. Perform economic analyses for new products, to evaluate technologies, and to assess business processes.
3. Prepare business plans that include financial details, marketing strategies, and design decisions based on target costs and forecasted rate of return on investment capital.
4. Combine technical, financial, marketing, human resources, and strategic considerations in decision making on engineering projects.

Learning Goals (LGs) Adopted in May 2025

1. Make decisions concerning product or process development.
2. Perform economic analyses for new products, to evaluate technologies, and to assess business processes.
3. Prepare business plans that include financial details, marketing and design decisions based on project requirements, project goals, quality assurance, scheduling, work breakdown structure, targeted profit and costs, and forecasted rate of return on investment capital.
4. Combine technical, financial, strategic considerations, and human resources such as project team formation, communication skills and personality dynamics, in decision making on engineering projects.

Meeting concluded at 1:30 pm.